

Course Project - Graduate Operating Systems - Fall 2026

Overview

A key component of Graduate Operating Systems will be the creation and / or modification of a system artifact for the purposes of instrumentation and analysis. The project itself will involve an exploration of the prevailing literature, identification of key challenges, development or improvement of a system, and a fully analysis of said system proving successful achievement of your system design goals. Alternatively stated, your goal is to execute a successful systems pilot project that could plausibly turn into a fully-fledged systems research paper.

The course project constitutes 40% of your course grade.

Projects

For your project, you must select from one of the following topics as an initial starting point for the research:

- *Open-source software/tool-box acceleration or functionality enhancement:* Take an existing software package/tool-box that is used by researchers at Notre Dame and explore key bottlenecks or performance constraints in the system. Explore how one might effectively distribute or compute across a reasonable set of computation nodes and measure/understand how such computation can scale.
 - Example: Explore how one might transition a particular piece of software onto a GPU
 - Example: Compare/contrast performance running on a laptop, desktop, and Raspberry Pi
- *Discrete event simulation:* Pick a particular focus topic from the course (storage, scheduling, virtual memory, virtualization) or something related to your research and create a discrete event simulator to conduct a "bake off" between various strategies.
- *System for learning and learning for system:* Write a survey paper by reading over 15 papers in top-tier system research venues that use machine learning techniques. A survey paper will be focused on a single topic. The flow of the survey paper should be: describe the research problem, discuss the surveyed papers, make comparison between them, identify their pros and cons, and point out possible future work.
- *Improvement paper:* Take a recent work from a top-tier venue in the system area. Attempt to improve the design/performance of the work.
- *Contemporary paper:* Take a recent work from a top-tier venue in the system area. Attempt to reproduce the results from the paper and evaluate whether or not the results are correct or are more nuanced.
- *Security:* While we do not cover much about system security in the class, you may i) take a recent security paper from top-tier venues (e.g. ACM CCS, IEEE

S&P, USENIX Security) and improve/reproduce it; or ii) choose a research topic of security and write a survey paper. You may choose papers that address security of machine learning or machine learning for security.

- *Bring your own project:* While it is possible to bring your own project, the project must be well-conceived and discussed with Prof. Kong in the concept phase. Note that your project may NOT be confined to OS but it needs to be system-related research, at least in a broad way.

Team

Your project may be completed in a group of two to four students. In line with neither conferences nor journals making special accommodations for group size, there will be no expectation of quality or quantity differences between groups of differing sizes. Hence, it is likely in your own interest to work in as large of a group size as you can plausibly manage.

However, please note that groups are fixed by the project proposal phase. It will also be in your interest to ensure that group members are adequately contributing to ensure the success of the group.

Deliverables

The deliverables for the course project are as follows:

Project Proposal	10%	1. A brief three slides set conveying the focus of your project and group members. Present orally (5~10 mins) in class. 2. An overview of your proposed course project taking special care to identify the key components, resources, and timeline to successfully achieve your work. The proposal should also properly contextualize the importance of your project with regards to the overall body of literature in systems research. A key portion is a statement of your research hypothesis(es) and how you intend to test said hypotheses. Submitted as a PDF.
Mid-term Progress Report	10%	1. You will be asked to make a presentation for your project in a talk of about 10 ~15 mins providing a status update / presentation to the whole class. 2. A report that summarizes your project progress. The report should include what you have done (such as challenges/problems you have identified and solved), and what you will continue to do to finish the project. Submitted as a PDF.

Final Paper	20%	1. You will be responsible for giving a final presentation of your project. The presentation will be about 15 ~ 20 mins long. 2. You will submit a final paper. The final paper will be a highly polished result within “striking distance” of submission to an academic conference. Submitted as a PDF.
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Tentative Dates

Project Proposal Presentation starts in the 4th week. Proposal PDF is due in one week after the presentation.

Mid-term Progress Report Presentation starts right after the mid-term break. The report PDF is due in one week after the presentation.

Final Paper Presentation starts in the second last week. The final paper is due in one week after the last class day.

Those dates are tentative and subject to change accordingly.

Submission of materials will be done via Canvas with all submission items allowing for unlimited resubmissions before the due date.